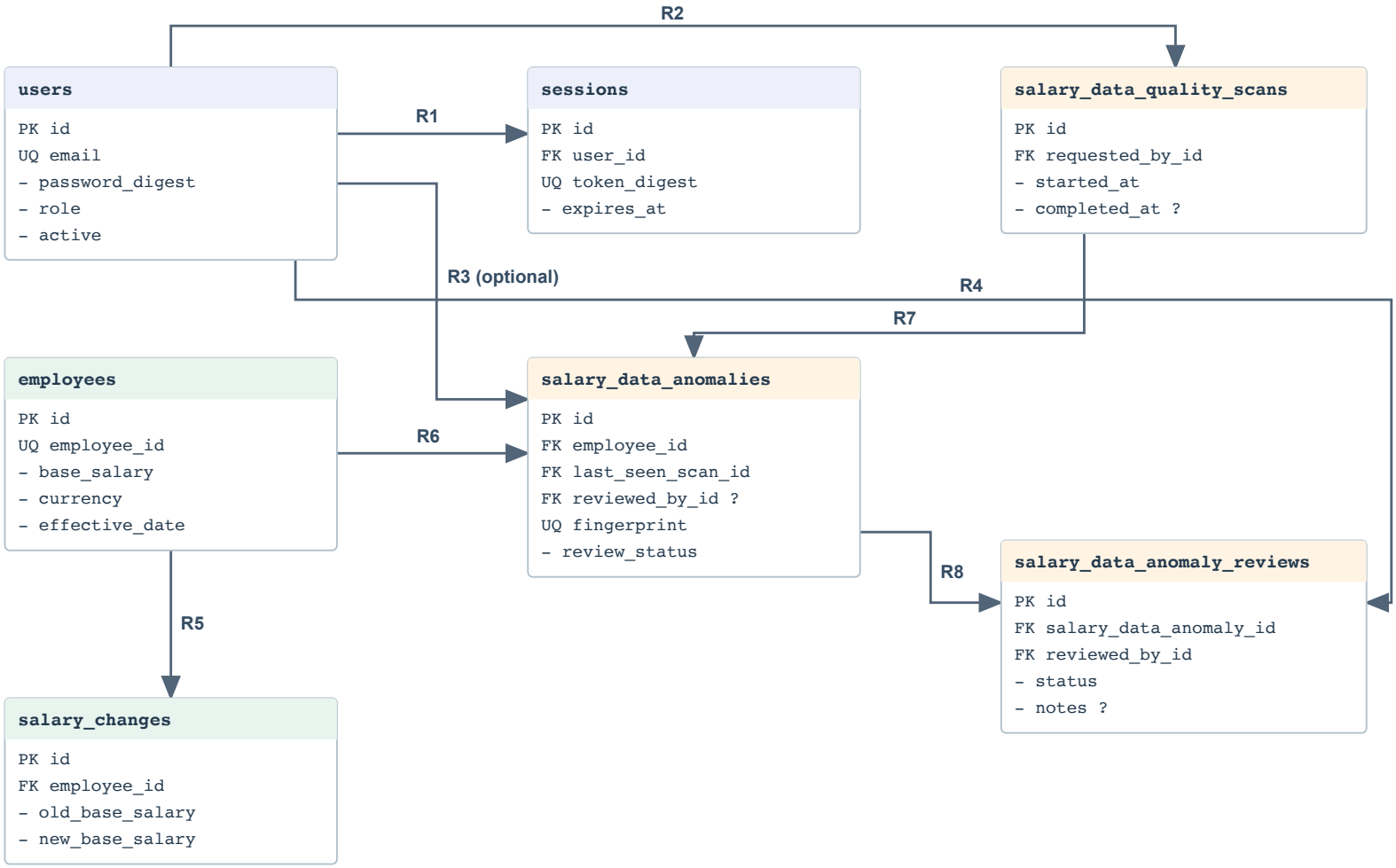
Employee Portal Database ERD

PostgreSQL | Schema version `20261013000004` | 2026-10-13



Key fields are shown above; full column definitions follow. PK = primary key, FK = foreign key, UQ = unique index, ? = nullable. Arrows point from parent to child; labels R1-R8 refer to the relationship dictionary on page 2.

Schema Overview and Relationships

7	72	8	19
Application tables	Columns	Foreign keys	Secondary indexes

This report documents `Backend/db/schema.rb`, verified against local development metadata on 2026-10-13. No employee records, password hashes, tokens or connection credentials were read or included. It does not certify the production database state.

Relationship Dictionary

Ref.	Parent	Child / Foreign Key	Child Requires Parent?
R1	`users.id`	`sessions.user_id`	Yes: exactly 1 parent
R2	`users.id`	`salary_data_quality_scans.requested_by_id`	Yes: exactly 1 parent
R3	`users.id`	`salary_data_anomalies.reviewed_by_id`	No: 0 or 1 parent
R4	`users.id`	`salary_data_anomaly_reviews.reviewed_by_id`	Yes: exactly 1 parent
R5	`employees.id`	`salary_changes.employee_id`	Yes: exactly 1 parent
R6	`employees.id`	`salary_data_anomalies.employee_id`	Yes: exactly 1 parent
R7	`salary_data_quality_scans.id`	`salary_data_anomalies.last_seen_scan_id`	Yes: exactly 1 parent
R8	`salary_data_anomalies.id`	`salary_data_anomaly_reviews.salary_data_anomaly_id`	Yes: exactly 1 parent

Every parent can have zero or many children. No foreign key is unique, so none of these relationships is one-to-one. `salary_data_anomalies.reviewed_by_id` is the only nullable foreign key.

Scope and Identity

- Every application table has a sequence-generated `bigint id` primary key and required `created_at` / `updated_at` timestamps.
- `employees.employee_id` is a string business ID. Other tables' numeric `employee_id` foreign keys reference `employees.id`, not that business ID.
- Framework bookkeeping tables `schema_migrations` and `ar_internal_metadata` are excluded from the seven-table count. Rails model `Compensation` is not a persisted table.
- This is the primary application schema. Production configuration also names cache, queue and cable databases; their schemas are not covered here.

employees

Current employee directory and annual base salary.

Column	SQL Type	NULL?	Default	Key / Reference
id	bigint	No	Sequence-generated	**PK**
employee_id	varchar	No	–	**UQ**
name	varchar	No	–	-
department	varchar	No	–	-
country	varchar	No	–	-
designation	varchar	No	–	-
status	varchar	No	–	-
base_salary	numeric(18,2)	No	–	-
currency	varchar	No	–	-
effective_date	date	No	–	-
created_at	timestamp(6)	No	–	-
updated_at	timestamp(6)	No	–	-

SQL type aliases: varchar = character varying; timestamp(6) = timestamp(6) without time zone. A dash under Default means no declared default, not that NULL is permitted. Sequence-generated IDs use PostgreSQL nextval defaults.

Indexes

`index_employees_on_base_salary_and_employee_id`
Non-unique btree (base_salary, employee_id)

`index_employees_on_country_and_department_and_status`
Non-unique btree (country, department, status)

`index_employees_on_department`
Non-unique btree (department)

`index_employees_on_employee_id`
Unique btree (employee_id)

`index_employees_on_status`
Non-unique btree (status)

Database Checks

`employees_nonnegative_salary`
base_salary must be nonnegative.

`employees_supported_currency`
currency must be INR, USD, GBP, EUR or JPY.

`employees_supported_status`
status must be active or inactive.

Application Behavior

employee_id is a unique business identifier, not a foreign key. Department, country and designation are strings, not separate lookup tables. Future effective dates and zero salaries are valid.

salary_changes

Old/new compensation values for confirmed salary changes.

Column	SQL Type	NULL?	Default	Key / Reference
id	bigint	No	Sequence-generated	**PK**
employee_id	bigint	No	–	**FK** employees.id
old_base_salary	numeric(18,2)	No	–	-
new_base_salary	numeric(18,2)	No	–	-
old_currency	varchar	No	–	-
new_currency	varchar	No	–	-
old_effective_date	date	No	–	-
effective_date	date	No	–	-
changed_at	timestamp(6)	No	–	-
created_at	timestamp(6)	No	–	-
updated_at	timestamp(6)	No	–	-

SQL type aliases: varchar = character varying; timestamp(6) = timestamp(6) without time zone. A dash under Default means no declared default, not that NULL is permitted. Sequence-generated IDs use PostgreSQL nextval defaults.

Indexes

`index_salary_changes_on_employee_id`
Non-unique btree (employee_id)

`index_salary_changes_on_employee_id_and_changed_at_and_id`
Non-unique btree (employee_id, changed_at, id)

Database Checks

`salary_changes_supported_currencies`
Both old and new currency must be INR, USD, GBP, EUR or JPY.

`salary_changes_nonnegative_salaries`
Old and new salary amounts must both be nonnegative.

Application Behavior

Salary updates and history insertion happen in one locked transaction. Persisted history is read-only through Rails, not through a database trigger. There is no changed_by user column. Initial salaries need not have a history row.

users

HR authentication identities, separate from payroll employees.

Column	SQL Type	NULL?	Default	Key / Reference
id	bigint	No	Sequence-generated	**PK**
email	varchar	No	–	**UQ**
name	varchar	No	–	-
role	varchar	No	superadmin	-
password_digest	varchar	No	–	-
active	boolean	No	true	-
created_at	timestamp(6)	No	–	-
updated_at	timestamp(6)	No	–	-

SQL type aliases: varchar = character varying; timestamp(6) = timestamp(6) without time zone. A dash under Default means no declared default, not that NULL is permitted. Sequence-generated IDs use PostgreSQL nextval defaults.

Indexes

`index_users_on_email`
Unique btree (email)

Database Checks

`users_superadmin_role`
role must be superadmin.

Application Behavior

Rails normalizes emails and uses bcrypt password digests. The schema allows multiple unique-email users; the application currently seeds one HR user. Only superadmin is supported. The email index is not case-insensitive by itself.

sessions

Server-side records for authenticated bearer sessions.

Column	SQL Type	NULL?	Default	Key / Reference
id	bigint	No	Sequence-generated	**PK**
user_id	bigint	No	–	**FK** users.id
token_digest	varchar	No	–	**UQ**
expires_at	timestamp(6)	No	–	-
created_at	timestamp(6)	No	–	-
updated_at	timestamp(6)	No	–	-

SQL type aliases: varchar = character varying; timestamp(6) = timestamp(6) without time zone. A dash under Default means no declared default, not that NULL is permitted. Sequence-generated IDs use PostgreSQL nextval defaults.

Indexes

index_sessions_on_expires_at
Non-unique btree (expires_at)

index_sessions_on_token_digest
Unique btree (token_digest)

index_sessions_on_user_id
Non-unique btree (user_id)

Database Checks

No CHECK constraints declared for this table.

Application Behavior

Only SHA-256 token digests are stored, not raw bearer tokens. Rails issues eight-hour sessions and checks user activity and expiry. Changing a password or activity status revokes sessions through application logic.

salary_data_quality_scans

Manual salary-check runs, authors, timings and aggregate counters.

Column	SQL Type	NULL?	Default	Key / Reference
id	bigint	No	Sequence-generated	**PK**
requested_by_id	bigint	No	–	**FK** users.id
started_at	timestamp(6)	No	–	-
completed_at	timestamp(6)	Yes	–	-
checked_employees	integer	No	0	-
detected_findings	integer	No	0	-
new_findings	integer	No	0	-
affected_employees	integer	No	0	-
cleared_findings	integer	No	0	-
created_at	timestamp(6)	No	–	-
updated_at	timestamp(6)	No	–	-

SQL type aliases: varchar = character varying; timestamp(6) = timestamp(6) without time zone. A dash under Default means no declared default, not that NULL is permitted. Sequence-generated IDs use PostgreSQL nextval defaults.

Indexes

index_salary_data_quality_scans_on_requested_by_id
Non-unique btree (requested_by_id)

Database Checks

No CHECK constraints declared for this table.

Application Behavior

Counters default to zero. The scan service uses batched reads and a transaction advisory lock. It does not edit employees. Scan counters are historical summaries; there is no per-scan finding-membership join table.

salary_data_anomalies

Persisted flags, detection evidence and the latest HR review state.

Column	SQL Type	NULL?	Default	Key / Reference
id	bigint	No	Sequence-generated	**PK**
employee_id	bigint	No	–	**FK** employees.id
last_seen_scan_id	bigint	No	–	**FK** salary_data_quality_scans.id
fingerprint	varchar	No	–	**UQ**
rule_code	varchar	No	–	-
severity	varchar	No	–	-
snapshot	jsonb	No	{}	-
observed	jsonb	No	{}	-
reference	jsonb	No	{}	-
active	boolean	No	true	-
review_status	varchar	No	pending	-
review_notes	text	Yes	–	-
reviewed_by_id	bigint	Yes	–	**FK** users.id
reviewed_at	timestamp(6)	Yes	–	-
cleared_at	timestamp(6)	Yes	–	-
created_at	timestamp(6)	No	–	-
updated_at	timestamp(6)	No	–	-

SQL type aliases: varchar = character varying; timestamp(6) = timestamp(6) without time zone. A dash under Default means no declared default, not that NULL is permitted. Sequence-generated IDs use PostgreSQL nextval defaults.

Indexes, constraints and review-state behavior continue on the next page.

salary_data_anomalies: Supporting Details

Indexes

`index_salary_anomalies_on_review_queue`
Non-unique btree (active, review_status, created_at)

`index_salary_data_anomalies_on_employee_id`
Non-unique btree (employee_id)

`index_salary_data_anomalies_on_fingerprint`
Unique btree (fingerprint)

`index_salary_data_anomalies_on_last_seen_scan_id`
Non-unique btree (last_seen_scan_id)

`index_salary_data_anomalies_on_reviewed_by_id`
Non-unique btree (reviewed_by_id)

Database Checks

`salary_anomalies_review_status`
review_status must be pending, investigating, confirmed_error or legitimate.

`salary_anomalies_severity`
severity must be warning or info.

Application Behavior

The fingerprint uniquely identifies matching evidence. JSONB stores the employee snapshot, observed values and reference thresholds. last_seen_scan_id points to the most recent scan detecting this evidence. Reviewed and cleared timestamps are optional.

salary_data_anomaly_reviews

Append-only HR review decisions and notes for each finding.

Column	SQL Type	NULL?	Default	Key / Reference
id	bigint	No	Sequence-generated	**PK**
salary_data_anomaly_id	bigint	No	–	**FK** salary_data_anomalies.id
reviewed_by_id	bigint	No	–	**FK** users.id
status	varchar	No	–	-
notes	text	Yes	–	-
created_at	timestamp(6)	No	–	-
updated_at	timestamp(6)	No	–	-

SQL type aliases: varchar = character varying; timestamp(6) = timestamp(6) without time zone. A dash under Default means no declared default, not that NULL is permitted. Sequence-generated IDs use PostgreSQL nextval defaults.

Indexes

`index_salary_data_anomaly_reviews_on_reviewed_by_id`
Non-unique btree (reviewed_by_id)

`index_salary_data_anomaly_reviews_on_salary_data_anomaly_id`
Non-unique btree (salary_data_anomaly_id)

Database Checks

`salary_anomaly_reviews_status`
status must be pending, investigating, confirmed_error or legitimate.

Application Behavior

Rails requires notes for confirmed_error and legitimate and limits notes to 2,000 characters. Persisted reviews are read-only through Rails; SQL can still change them. The anomaly row caches the latest decision while this table keeps the audit trail.

Database Guarantees and Operating Notes

Enforced by PostgreSQL

- Seven primary keys, eight foreign keys and 9 CHECK constraints. There are 19 secondary indexes, four unique, in addition to the seven primary-key indexes.
- Unique secondary indexes protect employee business IDs, user emails, bearer token digests and anomaly fingerprints.
- Annual salary amounts use exact `numeric(18,2)`: up to 16 integer digits and two decimal digits. Supported currencies are INR, USD, GBP, EUR and JPY. Monetary aggregates never combine currencies.
- Dates use PostgreSQL `date`. Timestamps use `timestamp(6) without time zone`; Rails manages application time interpretation. JSONB evidence defaults to an empty object and is required, but has no database-level JSON shape constraint.
- All eight foreign keys have default NO ACTION behavior for deletes and updates: no database ON DELETE CASCADE is declared.

Enforced by Rails, Not Database Triggers

- Required strings must be nonblank; NULL constraints alone do not reject whitespace. Email normalization and format checks run in Rails.
- Salary changes require explicit confirmation. Employee updates and their history rows are written atomically while locking the employee.
- Salary history and anomaly review records are read-only once persisted through Active Record. Direct SQL can bypass this application safeguard.
- Review notes are required for Confirmed error and Legitimate, and are limited to 2,000 characters. Those conditional requirements are not SQL constraints.
- Deleting an employee with history/findings, or a finding with reviews, is restricted by model associations. User deletion destroys sessions through Rails; audit foreign keys can still prevent deleting that user.

Data-Quality Lifecycle

A scan writes summary counters and fingerprints matching findings. Unchanged reruns retain the existing finding and HR decision. Changed evidence creates a new Pending finding; cleared evidence stays stored with `active=false` and `cleared_at`. HR decisions append review events, and the finding caches the latest reviewer, notes and status. Scans never automatically alter salary.

Deployment and Sources

Latest migration: `20261013000004_create_salary_data_quality_reviews.rb`. Apply migrations before deploying code that queries the review tables. The schema report includes these local changes; deployment is separate. See `Backend/db/schema.rb`, `Backend/db/migrate/`, `Backend/app/models/` and the salary-update/check/review services.

The adjacent JSON preserves exact PostgreSQL types, index definitions, foreign keys, default expressions and CHECK expressions. The SVG and Mermaid files are editable ERD sources. This report contains structure only, not live employee counts or private data.